# 2023
# VCE
# Specialist Mathematics
# Year 12
# Trial Examination 1

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# Victorian Certificate of Education
# 2023

**STUDENT NUMBER**

| | | | | | | | | | | Letter |
|---|---|---|---|---|---|---|---|---|---|---|
| Figures | | | | | | | | | | |
| Words | | | | | | | | | | |

# SPECIALIST MATHEMATICS

# Trial Written Examination 1

Reading time: 15 minutes
Total writing time: 1 hour

## QUESTION AND ANSWER BOOK

**Structure of book**

| *Number of questions* | *Number of questions to be answered* | *Number of marks* |
|---|---|---|
| 10 | 10 | 40 |

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination room: any technology (calculators or software), notes of any kind, blank sheets of paper and/or white out liquid/tape.

**Materials supplied**

- Question and answer book of 22 pages with a detachable sheet of miscellaneous formulas at the end of this booklet.

**Instructions**

- Detach the formula sheet from the end of this book during reading time.
- Write your **student number** in the space provided above on this page.
- All written responses must be in English.

**Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.**

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**Instructions**

Answer **all** questions in the spaces provided.
Unless otherwise specified an **exact** answer is required to a question.
In questions where more than one mark is available, appropriate working **must** be shown.
Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
Take the **acceleration due to gravity** to have magnitude $g$ m/s$^2$, where $g = 9.8$.

**Question 1** (4 marks)

Solve the equation $z^4 + 16 = 0,\ z \in C,$ giving your answers in rectangular $a + bi$ form.

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**Question 2** (3 marks)

Solve the differential equation $\frac{dy}{dx} = \frac{x\left(9+4y^2\right)}{6}$ given $y(0) = \frac{3}{2}$.

Express your answer in the form $y = f(x)$.

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**Question 3** (3 marks)

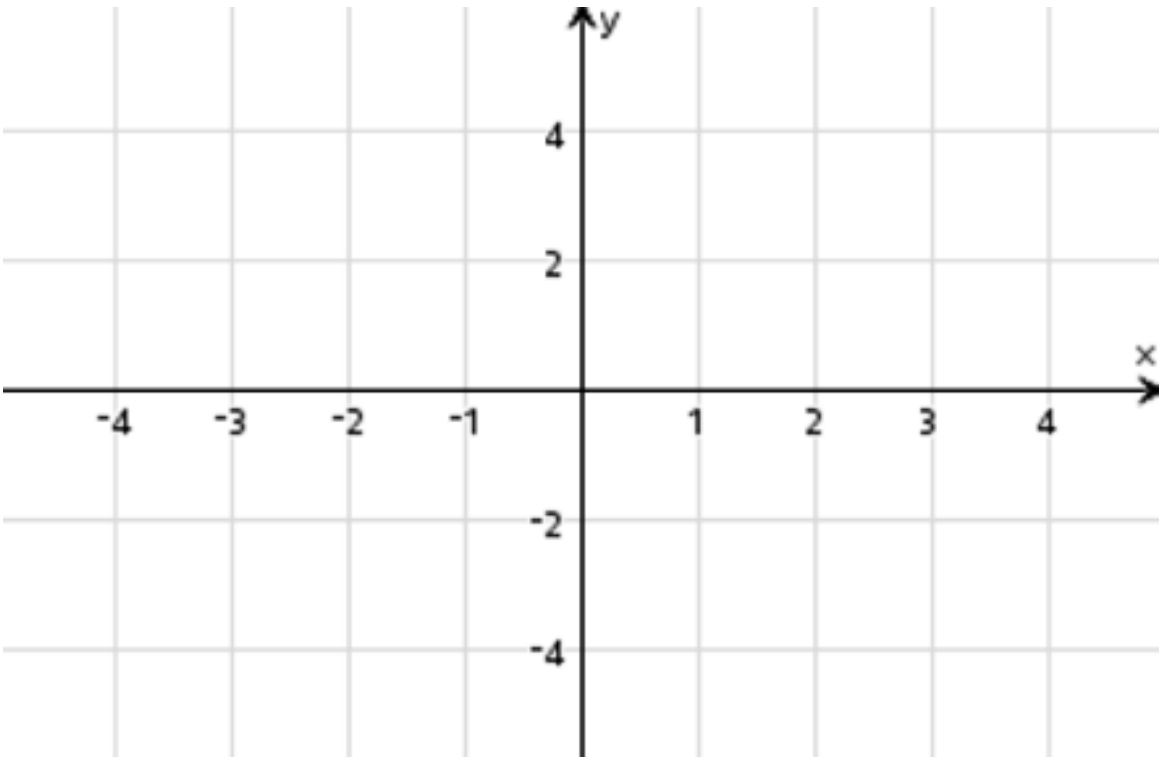
Let $f : D \to R,\ f(x) = \dfrac{x}{x^2 - 4}$, where $D$ is the maximal domain of $f$.

**i.** Find $f'(x)$ and hence show that the graph of $f$ has no turning points.

1 mark

**ii.** Given that $f''(x) = \dfrac{2x(x^2 + 12)}{(x^2 - 4)^3}$ find the coordinates of any points of inflexion and sketch the graph of $f(x) = \dfrac{x}{x^2 - 4}$ on the axes below, labelling the equations of all asymptotes.

2 marks

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**Question 4** (3 marks)

A particle moves in a straight line. At a time $t$ seconds, the particle has a displacement of $x$ metres and a velocity of $v \text{ ms}^{-1}$ and an acceleration $a \text{ ms}^{-2}$. Initially the particle is at the origin and has a velocity of $2 \text{ ms}^{-1}$. If the acceleration $a = -\frac{4}{3}e^{-\frac{2x}{3}}$ and the velocity of the particle is always positive.

**a.** Show that $v = 2e^{-\frac{x}{3}}$

1 mark

**b.** After three seconds, the particle has a displacement of $\log_e(s)$, find the value of $s$.

2 marks

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**Question 5** (3 marks)

The weights of a piece of corn are normally distributed with a mean of 100 gm with a variance of 12 gm$^2$. $Z$ has the standard normal distribution and given that $\Pr(Z < 1.5) = 0.933$.

**a.** Determine the probability that the total weight of three independent pieces of corn will be between 291 and 300 gms. Give your answer correct to three decimal places.

2 marks

**b.** An **approximate** 95% confidence interval for $n$ pieces of corn was found to be 99 to 101 gm. Given that $\Pr(-2 < Z < 2) = 0.95$, where Z is a standard normal random variable.

Determine the value of $n$.

1 mark

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**Question 6** (5 marks)

On a plantation in the year 2000 there were initially 50 kangaroos. The population growth of the number of kangaroos $K$ on the plantation is given by $\frac{dK}{dt} = \frac{K}{5}\left(1 - \frac{K}{500}\right)$, where $t$ is the time in years after 2000.

**a.** Solve the differential equation to find $K$ in terms of $t$.

4 marks

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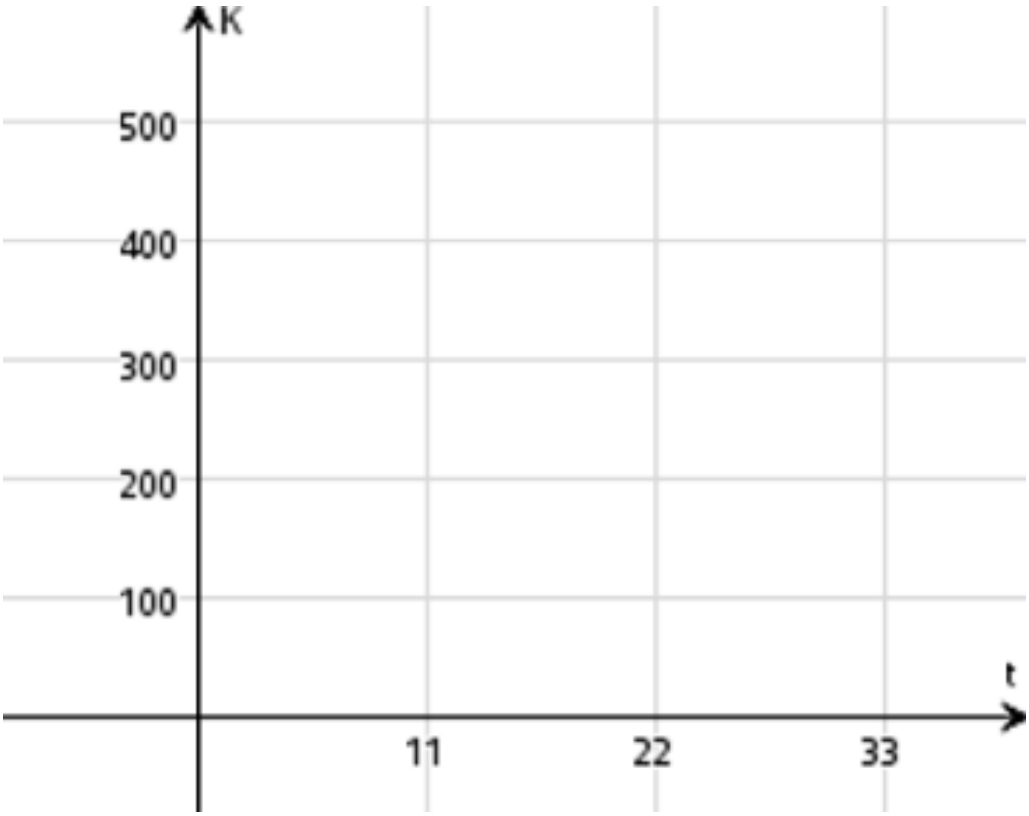
**b.** It is known that the number of kangaroos on the plantation was increasing most rapidly around the year 2011. Sketch the graph of $K$ versus $t$ on the axes below, clearly labelling the point of inflexion, the equations of any asymptotes and coordinates of any axial intercepts.

1 mark

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**Question 7** (4 marks )

When the area bounded by the curve the curve $y=\sqrt{3x+4}$ the $x$-axis $x=0$ and $x=2$ is rotated about the $x$-axis it forms a volume of revolution, find the surface area of this volume of revolution.

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**Question 8** (4 marks )

Prove by induction that $5^{2n}+3n-1$ is divisible by 9 for $n \in N$.

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**Question 9** (6 marks)

A plane has the vector equation $\underset{\sim}{r}(s,t)=(3+4t+3s)\underset{\sim}{i}+(-2+t+2s)\underset{\sim}{j}+(1+2t-s)\underset{\sim}{k}$

**a.** Show that the equation of the plane can be written in the form $x-2y-z=6$.

2 marks

**b.** One line has the vector equation $\underset{\sim}{r}(t)=2\underset{\sim}{i}-\underset{\sim}{j}+z_0\underset{\sim}{k}+t\left(\underset{\sim}{i}-2\underset{\sim}{j}+c\underset{\sim}{k}\right)$, find the values of $z_0$ and $c$ for which the line and the plane do not intersect.

2 marks

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**c.** Another line has the equation $\frac{x-3}{2}=\frac{y+1}{b}=z-2$, this line intersects the plane making an angle of $30^0$ with the plane. Determine the value(s) of $b$.

2 marks

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**Question 10** (5 marks)

Let $I_n = \int_0^1 x^{2n+1} e^{x^2}\, dx$ for integers $n \geq 0$.

**a.** Show that $I_0 = \frac{1}{2}(e-1)$

1 mark

**b.** Using integration by parts on $I_n$, show that $I_{n+1} = \frac{e}{2} - (n+1) I_n$

3 marks

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**c.** Hence evaluate $I_3$.

1 mark

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**EXTRA WORKING SPACE**

**End of question and answer book for the**
**2023 Kilbaha VCE Specialist Mathematics Trial Examination 1**



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# SPECIALIST MATHEMATICS

# Written examination 1

## FORMULA SHEET

### Directions to students

| |
|---|
| Detach this formula sheet during reading time.<br><br>This formula sheet is provided for your reference. |

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# Specialist Mathematics formulas

## Mensuration

| | | | |
|---|---|---|---|
| area of a circle segment | $\frac{r^2}{2}(\theta - \sin(\theta))$ | volume of a sphere | $\frac{4}{3}\pi r^3$ |
| volume of a cylinder | $\pi r^2 h$ | area of a triangle | $\frac{1}{2}bc\sin(A)$ |
| volume of a cone | $\frac{1}{3}\pi r^2 h$ | sine rule | $\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$ |
| volume of a pyramid | $\frac{1}{3}Ah$ | cosine rule | $c^2 = a^2 + b^2 - 2ab\cos(C)$ |

## Algebra, number and structure ( complex numbers )

| | | |
|---|---|---|
| $z = x + yi = r(\cos(\theta) + i\sin(\theta)) = r\,\text{cis}(\theta)$ | $\lvert z\rvert = \sqrt{x^2 + y^2} = r$ | |
| $-\pi < \text{Arg}(z) \le \pi$ | $z_1 z_2 = r_1 r_2\,\text{cis}(\theta_1 + \theta_2)$ | |
| $\frac{z_1}{z_2} = \frac{r_1}{r_2}\,\text{cis}(\theta_1 - \theta_2)$ | de Moivre's theorem | $z^n = r^n\text{cis}(n\theta)$ |

## Circular (trigonometric) functions

| | |
|---|---|
| $\cos^2(x) + \sin^2(x) = 1$ | |
| $1 + \tan^2(x) = \sec^2(x)$ | $\cot^2(x) + 1 = \text{cosec}^2(x)$ |
| $\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y)$ | $\sin(x-y) = \sin(x)\cos(y) - \cos(x)\sin(y)$ |
| $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$ | $\cos(x-y) = \cos(x)\cos(y) + \sin(x)\sin(y)$ |
| $\tan(x+y) = \frac{\tan(x) + \tan(y)}{1 - \tan(x)\tan(y)}$ | $\tan(x-y) = \frac{\tan(x) - \tan(y)}{1 + \tan(x)\tan(y)}$ |
| $\sin(2x) = 2\sin(x)\cos(x)$ | |
| $\cos(2x) = \cos^2(x) - \sin^2(x)$ <br> $= 2\cos^2(x) - 1 = 1 - 2\sin^2(x)$ | $\tan(2x) = \frac{2\tan(x)}{1 - \tan^2(x)}$ |
| $\sin^2(ax) = \frac{1}{2}(1 - \cos(2ax))$ | $\cos^2(ax) = \frac{1}{2}(1 + \cos(2ax))$ |

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## Data analysis, probability and statistics

| | | |
|---|---|---|
| for independent random variables $X_1, X_2 .... X_n$ | $E(aX_1+b)=aE(X_1)+b$<br>$E(a_1X_1+a_2X_2+...a_nX_n)$<br>$=a_1E(X_1)+a_2E(X_2)+...+a_nE(X_n)$ | |
| | $\text{Var}(aX_1+b)=a^2\text{Var}(X_1)$<br>$\text{Var}(a_1X_1+a_2X_2+...a_nX_n)$<br>$=a_1^2\text{Var}(X_1)+a_2^2\text{Var}(X_2)+...+a_n^2\text{Var}(X_n)$ | |
| for independent identically distributed variables $X_1, X_2 .... X_n$ | $E(X_1+X_2+...+X_n)=n\mu$ | |
| | $\text{Var}(X_1+X_2+...+X_n)=n\sigma^2$ | |
| approximate confidence interval for $\mu$ | $\left(\bar{x}-z\frac{s}{\sqrt{n}}, \bar{x}+z\frac{s}{\sqrt{n}}\right)$ | |
| distribution of sample mean $\bar{X}$ | mean | $E(\bar{X})=\mu$ |
| | variance | $\text{Var}(\bar{X})=\frac{\sigma^2}{n}$ |

## Vectors in two and three dimensions

| | |
|---|---|
| $\underset{\sim}{r}(t)=x(t)\underset{\sim}{i}+y(t)\underset{\sim}{j}+z(t)\underset{\sim}{k}$ | $\lvert\underset{\sim}{r}(t)\rvert=\sqrt{x(t)^2+y(t)^2+z(t)^2}$ |
| | $\dot{\underset{\sim}{r}}(t)=\frac{d\underset{\sim}{r}}{dt}=\frac{dx}{dt}\underset{\sim}{i}+\frac{dy}{dt}\underset{\sim}{j}+\frac{dz}{dt}\underset{\sim}{k}$ |
| for $\underset{\sim}{r_1}=x_1\underset{\sim}{i}+y_1\underset{\sim}{j}+z_1\underset{\sim}{k}$ and $\underset{\sim}{r_2}=x_2\underset{\sim}{i}+y_2\underset{\sim}{j}+z_2\underset{\sim}{k}$ | vector scalar product<br>$\underset{\sim}{r_1}\cdot\underset{\sim}{r_2}=\lvert r_1\rvert\lvert r_2\rvert\cos(\theta)=x_1x_2+y_1y_2+z_1z_2$ |
| | vector cross product<br>$\underset{\sim}{r_1}\times\underset{\sim}{r_2}=\begin{vmatrix}\underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k}\\ x_1 & y_1 & z_1\\ x_2 & y_2 & z_2\end{vmatrix}=(y_1z_2-y_2z_1)\underset{\sim}{i}+(x_2z_1-x_1z_2)\underset{\sim}{j}+(x_1y_2-x_2y_1)\underset{\sim}{k}$ |
| vector equation of a line | $\underset{\sim}{r}(t)=\underset{\sim}{r_1}+t\underset{\sim}{r_2}=(x_1+x_2t)\underset{\sim}{i}+(y_1+y_2t)\underset{\sim}{j}+(z_1+z_2t)\underset{\sim}{k}$ |
| parametric equation of line | $x(t)=x_1+x_2t \quad y(t)=y_1+y_2t \quad z(t)=z_1+z_2t$ |
| vector equation of a plane | $\underset{\sim}{r}(s,t)=\underset{\sim}{r_0}+s\underset{\sim}{r_1}+t\underset{\sim}{r_2}$<br>$=(x_0+x_1s+x_2t)\underset{\sim}{i}+(y_0+y_1s+y_2t)\underset{\sim}{j}+(z_0+z_1s+z_2t)\underset{\sim}{k}$ |
| parametric equation of a plane | $x(s,t)=x_0+x_1s+x_2t \quad y(s,t)=y_0+y_1s+y_2t \quad z(s,t)=z_0+z_1s+z_2t$ |
| Cartesian equation of a plane | $ax+by+cz=d$ |

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## Calculus

| | |
|---|---|
| $\frac{d}{dx}\left(x^n\right)=nx^{n-1}$ | $\int x^n dx=\frac{1}{n+1}x^{n+1}+c\ ,\ n\neq -1$ |
| $\frac{d}{dx}\left(e^{ax}\right)=ae^{ax}$ | $\int e^{ax}dx=\frac{1}{a}e^{ax}+c$ |
| $\frac{d}{dx}\left(\log_e(x)\right)=\frac{1}{x}$ | $\int \frac{1}{x}dx=\log_e\left(\lvert x\rvert\right)+c$ |
| $\frac{d}{dx}\left(\sin(ax)\right)=a\cos(ax)$ | $\int \sin(ax)\,dx=-\frac{1}{a}\cos(ax)+c$ |
| $\frac{d}{dx}\left(\cos(ax)\right)=-a\sin(ax)$ | $\int \cos(ax)\,dx=\frac{1}{a}\sin(ax)+c$ |
| $\frac{d}{dx}\left(\tan(ax)\right)=a\sec^2(ax)$ | $\int \sec^2(ax)\,dx=\frac{1}{a}\tan(ax)+c$ |
| $\frac{d}{dx}\left(\cot(ax)\right)=-a\,\mathrm{cosec}^2(ax)$ | $\int \mathrm{cosec}^2(ax)\,dx=-\frac{1}{a}\cot(ax)+c$ |
| $\frac{d}{dx}\left(\sec(ax)\right)=a\sec(ax)\tan(ax)$ | $\int \sec(ax)\tan(ax)\,dx=\frac{1}{a}\sec(ax)+c$ |
| $\frac{d}{dx}\left(\mathrm{cosec}(ax)\right)=-a\,\mathrm{cosec}(ax)\cot(ax)$ | $\int \mathrm{cosec}(ax)\cot(ax)\,dx=-\frac{1}{a}\mathrm{cosec}(ax)+c$ |
| $\frac{d}{dx}\left(\sin^{-1}(ax)\right)=\frac{1}{\sqrt{1-(ax)^2}}$ | $\int \frac{1}{\sqrt{a^2-x^2}}dx=\sin^{-1}\left(\frac{x}{a}\right)+c,\ a>0$ |
| $\frac{d}{dx}\left(\cos^{-1}(ax)\right)=\frac{-1}{\sqrt{1-(ax)^2}}$ | $\int \frac{-1}{\sqrt{a^2-x^2}}dx=\cos^{-1}\left(\frac{x}{a}\right)+c,\ a>0$ |
| $\frac{d}{dx}\left(\tan^{-1}(ax)\right)=\frac{a}{1+(ax)^2}$ | $\int \frac{a}{a^2+x^2}dx=\tan^{-1}\left(\frac{x}{a}\right)+c$ |
| | $\int (ax+b)^n dx=\frac{1}{a(n+1)}(ax+b)^{n+1}+c,\ n\neq -1$ |
| | $\int (ax+b)^{-1}dx=\frac{1}{a}\log_e\left(\lvert ax+b\rvert\right)+c$ |

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## Calculus- continued

| | |
|---|---|
| product rule | $\frac{d}{dx}(uv)=u\frac{dv}{dx}+v\frac{du}{dx}$ |
| quotient rule | $\frac{d}{dx}\left(\frac{u}{v}\right)=\frac{v\frac{du}{dx}-u\frac{dv}{dx}}{v^2}$ |
| chain rule | $\frac{dy}{dx}=\frac{dy}{du}\frac{du}{dx}$ |
| integration by parts | $\int u\frac{dv}{dx}dx=uv-\int v\frac{du}{dx}dx$ |
| Euler's method | If $\frac{dy}{dx}=f(x),\ x_0=a$ and $y_0=b,$ then $x_{n+1}=x_n+h$ and $y_{n+1}=y_n+hf\left(x_n,y_n\right)$ |
| arc length parametric | $\int_{t_1}^{t_2}\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$ |
| surface area Cartesian about the *x*-axis | $\int_{x_1}^{x_2}2\pi y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$ |
| surface area Cartesian about the *y*-axis | $\int_{y_1}^{y_2}2\pi x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$ |
| surface area parametric about the *x*-axis | $\int_{t_1}^{t_2}2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$ |
| surface area parametric about the *y*-axis | $\int_{t_1}^{t_2}2\pi x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$ |

## Kinematics

| | | |
|---|---|---|
| acceleration | $a=\frac{d^2x}{dt^2}=\frac{dv}{dt}=v\frac{dv}{dx}=\frac{d}{dx}\left(\frac{1}{2}v^2\right)$ | |
| constant acceleration formulas | $v=u+at$ | $s=ut+\frac{1}{2}t^2$ |
| | $v^2=u^2+2as$ | $s=\frac{1}{2}(u+v)t$ |

**END OF FORMULA SHEET**

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